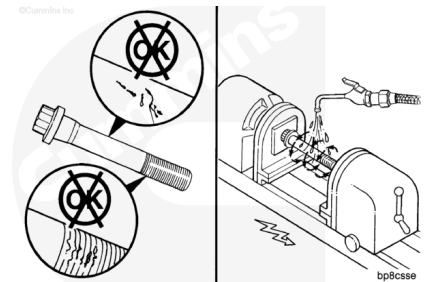


Trainings ▾

Magnetic Crack Inspect

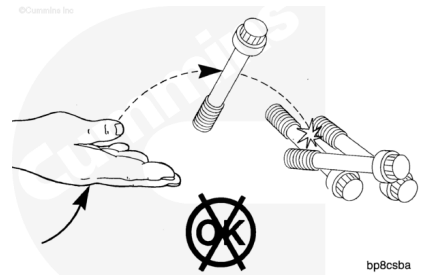
This procedure describes the visual and magnetic particle inspection of special and high strength capscrews for the:

- Cylinder head
- Main bearing cap
- Flywheel
- Connecting rod
- Crankshaft pulley or adapter
- Lifting bracket
- Crankshaft counterweight.



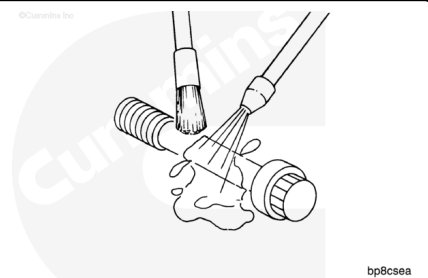
⚠ CAUTION ⚠

Prevent damage to the capscrews. Nicks in the body of the capscrew can cause an area of stress that can fail during engine operation. Damage to the threads will cause torque values to be incorrect and will damage the mating parts.



⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

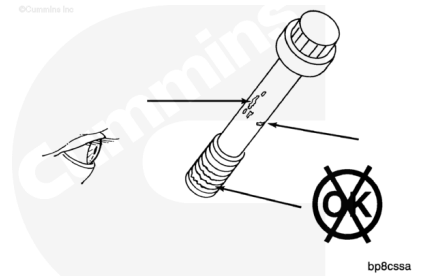
If necessary, use a wire wheel to remove all rust, corrosion, and dirt from the capscrews.

Use solvent to clean the capscrew and dry with compressed air.

Check the capscrew. The capscrew **must** be replaced if:

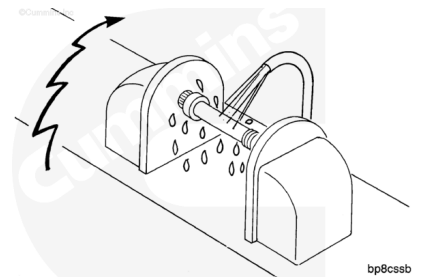
- Threads are damaged
- Rust or corrosion has caused pitting in the body
- The body is nicked, galled, bent, or stretched.

Some capscrews (cylinder head, main bearing cap, connecting rods) have threads that have been formed by a rolling operation during manufacture. Repair of rolled threads by use of a thread die is **not** recommended. The thread die can create a sharp corner on the minor diameter (root) of the threads. This sharp corner can cause an area of increased stress.



Use a magnetic particle testing machine such as Magnaflux.

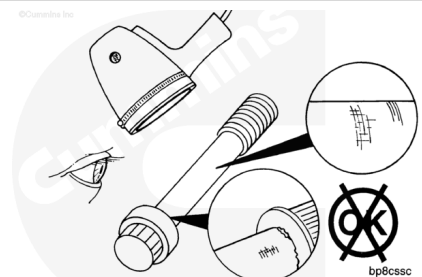
Use the continuous method. Apply a head shot of 300 to 400 amperes VDC or rectified VAC.



Use an ultraviolet light. Check for indications of cracks.

The magnetic particles tend to form on sharp corners and edges. Do **not** mistake these for cracks.

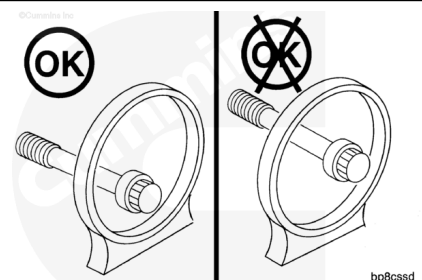
Any indications of cracks are **not** acceptable.



Prepare the machine for a coil shot.

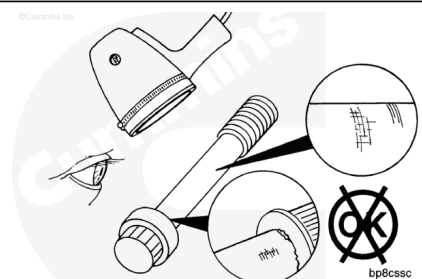
Make sure the capscrew is near one side of the coil and **not** in the center.

Apply 1000 to 1350 ampere-turns.



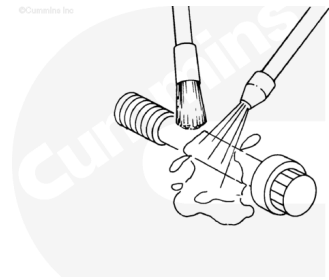
Use an ultraviolet light. Check for indications of cracks.

Any indications of cracks are **not** acceptable.



⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Demagnetize the capscrew thoroughly.

Use solvent to clean the capscrew and dry with compressed air.